
Position: The Logical Next Step for Efficient AI

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Abstract

Efficient inference is central to modern machine learning, from small on-device systems to centrally hosted large language models. However, current approaches rely on expensive linear algebraic computations, leading to ever-growing energy and deployment costs. This position paper argues for the necessity of radical hardware-software co-design that directly learns computational primitives, unlocking substantial gains in efficiency. Rather than requiring exotic hardware like neuromorphic chips or quantum computers, this capitalizes on the scalability of mature digital hardware. Logic neural networks (LNNs) represent a promising direction by directly learning combinations of low-level logic operations, aligning naturally with transistor-based computation. We view the current limitations of LNN model capabilities as symptoms of a young and fragmented research ecosystem, not as fundamental barriers. To realize the full potential of LNNs, we call on the community to prioritize accessibility through open-source development and closer collaboration between hardware and machine learning research.

1 Introduction

Over the past decade, machine learning (ML) has seen extraordinary success, driven by the development of increasingly large models [1]. This expansion has led to rapidly increasing energy consumption, with inference emerging as the dominant cost as models are executed billions of times over their lifetimes. Meta and Google estimate that their inference accounts for 60-70% of their ML energy consumption [2, 3] and AWS¹ and Nvidia² estimate that their customer’s ML inference is responsible for 80-90% of their ML cost. This asymmetry is likely to intensify as AI becomes more pervasive [4]. Beyond technical considerations, societal pressures are also mounting. A recent poll³ suggests that the American population is more concerned about the environmental impact of AI than that of the meat or aviation industry.

Efficient inference is not only a concern at hyperscale, but a fundamental requirement in resource-constrained regimes where scaling is impractical. In many real-world settings, models must operate under strict limits on latency, memory, and power: keyword spotting to wearable health monitoring and must operate continuously within microwatt power budgets and kilobyte memory footprints [5], autonomous driving systems must make decisions in real-time [6], data selection systems in particle physics experiments must process data at rates approaching petabytes per second [7] or occupy minimal silicon area [8]. In these domains, inefficiencies cannot be amortized away by scale, nor

¹Amazon EC2 Update. 2026-01-15

²Nvidia’s T4 GPUs for AI Inferencing. 2026-01-15

³The Center for Public Affairs Research. 2026-01-15

mitigated by additional compute. Instead, inference efficiency directly determines feasibility and adoption.

Despite the urgency of efficient inference, the focus of the research community does not seem to reflect this. As shown in Figure 1, the fraction of papers accepted to ICML and NeurIPS that focus on efficient inference remains low. Both scaling and inference used to be some of the main topics in the field, but this has been displaced by other popular topics such as LLMs. While the fraction of papers looking at scaling has gone down, efficient inference has stayed fairly flat over the years. Following the increase in topics such as LLMs, we argue that scaling should become a much more important topic, which should be reflected by the papers in the field.

Most current efforts toward efficient ML focus on model compression techniques such as pruning [9], quantization [10], and knowledge distillation [11]. Despite their success [12], these methods do not address a core inefficiency: models are typically expressed as large-scale linear algebra operations but executed on transistor-based digital hardware. This mismatch between algorithmic intent and physical execution introduces significant computational overhead. While modern accelerators—highly optimized for massively parallel matrix operations—have delivered substantial efficiency gains, the slowing of Moore’s law due to physical limits suggests that continued improvement through hardware scaling alone is becoming unsustainable [13].

This fundamental challenge has motivated growing interest in alternative hardware paradigms that aim to reduce data movement, memory footprint, and energy consumption by rethinking how computation is performed. Approaches such as analog [14], in-memory [15], neuromorphic [16], photonic [17], and quantum computing [18] promise tighter coupling between computation and physical substrates. However, despite their conceptual appeal, these paradigms remain constrained by hardware immaturity and limited scalability, as they cannot yet be fabricated or deployed at the scale required for widespread adoption.

The linear-algebra surrogate underlying most modern ML inference can, in principle, be eliminated on von Neumann hardware, enabling step-change improvements in computational efficiency, while continuing to leverage the maturity and scalability of transistor-based hardware. Logic-based neural networks (LNNs) offer a concrete realization of this vision by expressing computation directly through Boolean logic, allowing fast and energy-efficient inference on conventional digital platforms and achieving state-of-the-art performance in fast-ML regimes [19–21]. **We argue that the logical next step in efficient AI inference is to align model architectures with the native capabilities of digital hardware, which requires a unified research effort on logic-based neural networks bridging the hardware and ML communities.** Although current LNN approaches remain limited in scale and tooling maturity [22], we interpret these limitations not as fundamental barriers but as symptoms of a fragmented research ecosystem that has yet to converge on shared abstractions, development tools, and evaluation standards. It is noteworthy that anecdotal evidence points to successful attempts at scaling to larger models, but no public code or documentation is available⁴.

To ensure progress, we need to lower the barrier to entry while ensuring reproducibility and collaboration. We propose to achieve this through the following steps:

- **Open-source toolchains:** Low-barrier frameworks for training LNNs and synthesis tools that enable hardware-realistic evaluation without specialized engineering expertise.
- **Hardware-aware benchmarks:** Standardized evaluation protocols that jointly report accuracy and inference cost (latency, energy, area) on shared hardware targets.

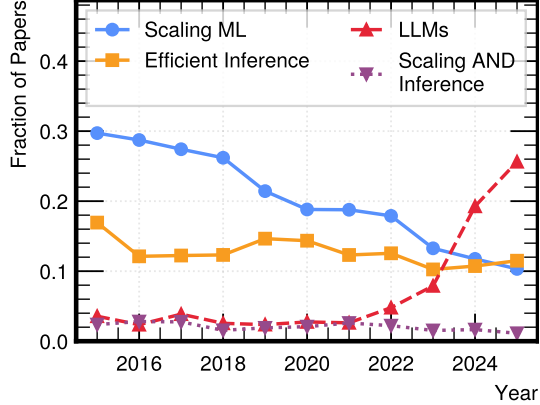


Figure 1: Fraction of total accepted abstracts at ICML and NeurIPS whose main topic is about scaling, efficient inference, or LLMs. This was scored with an LLM as a judge, and methodology details can be found in App. A.

⁴Ubiquitous Ultra-Efficient Mobile Video Super-resolution with Logic Gates, Felix Petersen, Louis Le Coeur. 2026-01-28

- **Cross-community collaboration:** Closer integration between ML, computer architecture, and digital design research to facilitate hardware–software co-design.

Without such structural changes, the field will continue to reward marginal accuracy gains while overlooking orders-of-magnitude improvements in inference efficiency.

2 Status Quo of Efficient Machine Learning

For three decades, the computing industry enjoyed automatic performance gains through Dennard scaling, allowing clock frequencies to double every 18 months without increasing power consumption [23]. As this came to an end, CPU clock speeds plateaued and the industry pivoted to parallelism. This shift continues today, but between 2006 and 2021, top GPU price-performance (FLOPS/\$) doubled only every 2.9 years⁵. This leaves a stark implication: if the computational paradigm remains unchanged, efficiency gains must also come from algorithmic and architectural innovation rather than exclusively waiting for better hardware. There is therefore extensive effort to make neural nets smaller through compression, and to challenge the current computing paradigms altogether.

2.1 Compression-Based Optimization

Compression based optimization like quantization, pruning, and knowledge distillation is standard practice for reducing inference costs. However, these methods share a fundamental limitation: they typically optimize within the linear algebra-based neural network paradigm rather than questioning whether that paradigm is appropriate for inference-constrained deployment.

Quantization reduces computational complexity by lowering the numerical precision of model parameters and arithmetic operations. The choice of the quantization scheme seeks to balance accuracy and hardware efficiency. Quantization can be applied post-training or incorporated during training, but in either case it primarily reduces the cost per operation rather than the number of operations. Even aggressively quantized models still require billions of multiply–accumulate operations per inference pass, and pushing precision below 8 bits often incurs a 3–10% accuracy degradation without extensive calibration or architectural support [24].

Pruning removes parameters or connections deemed redundant, either during training or as a post-training procedure, with the goal of reducing model size and inference cost. Although pruning reduces parameter count, retraining the induced architectures from scratch often yields performance matching or exceeding that of the original sparse models [25]. This suggests that pruning primarily serves as a mechanism for architecture discovery rather than parameter optimization. Consequently, much of the efficiency gain arises from identifying a more efficient architecture, rather than from sparsifying a given one.

Knowledge distillation trains a smaller *student* model to match the behavior of a larger *teacher* model, transferring performance into a more compact architecture. Distillation can be performed sequentially or jointly, with teacher and student trained simultaneously, but its impact on inference efficiency stems entirely from the student’s architecture. The teacher does not contribute to inference-time computation, and distillation does not directly reduce operation count or precision; instead, it enables smaller models to approximate the accuracy of larger ones while inheriting the inference characteristics of their chosen design [26].

2.2 The Matrix Multiplication Bottleneck

The compression-based methods mentioned above have delivered impressive gains, but they still rely on multiply–accumulate (MAC) operations as the fundamental unit of computation. In deployment, these map to binary logic on-chip, incurring substantial overhead through a surplus of low-level operations. While large-scale GPU inference is often memory-bound [27], in hardware-native settings such as FPGAs and custom ASICs, the dominant constraint shifts to the number of required logic operations, which directly translate to the transistor count, and ultimately chip area⁶. Thus, the mismatch between MAC-centric computation and transistor-level execution remains a key bottleneck in achieving truly efficient AI inference.

⁵Trends in GPU price-performance. 01-19-2026

⁶Taalas claim 10x inference speedups by deploying quantized LLMs on ASICs, avoiding memory movement.

2.3 Alternative Computing Paradigms

The limitations of compression-based optimization within dense, linear-algebra-based models have motivated substantial interest in alternative computing paradigms that seek to reduce inference cost by rethinking the physical substrate of computation itself. Rather than accelerating matrix multiplication, these approaches aim to bypass it by minimizing data movement, exploiting device physics, or encoding computation directly in hardware. Prominent examples include quantum computing, neuromorphic systems, and analog or in-memory computing.

Quantum ML promises asymptotic speedups for specific classes of problems by exploiting quantum superposition and entanglement [28]. However, current quantum hardware remains severely resource-constrained, noisy, and difficult to program. Near-term quantum devices support only shallow circuits with limited qubit counts, restricting practical ML applications to toy models [29]. Moreover, the lack of fault tolerance and the overhead of error correction place quantum ML far from deployment-relevant inference workloads, particularly in latency- or energy-constrained settings [30, 31].

Neuromorphic computing seeks to emulate biological neural systems using event-driven, spiking architectures that offer potentially large gains in energy efficiency [32]. Existing chips demonstrate impressive performance on tightly scoped tasks [33, 34], yet neuromorphic ML is still unusable in real-world tasks. Training spiking neural networks at scale remains difficult, toolchains are immature, and benchmarking against conventional models is inconsistent [35, 36]. As a result, neuromorphic approaches require problem reformulation and bespoke pipelines, limiting their accessibility and reproducibility.

Analog and in-memory computing architectures aim to reduce the dominant cost of ML inference—data movement—by performing computation directly where data resides [37]. Crossbar arrays using resistive memory or analog charge accumulation can, in principle, perform matrix-vector multiplication with high energy efficiency [38]. However, device non-idealities, noise, limited precision, and drift introduce substantial challenges for reliability and programmability [39]. In practice, these systems rely on correction and calibration layers that reintroduce complexity and erode theoretical efficiency gains [40, 41].

Manufacturing Maturity Beyond their weak integration with existing ML ecosystems, these alternative paradigms share a common barrier: they require manufacturing processes that do not yet exist at scale. Quantum systems need cryogenic environments at millikelvin temperatures [42]; memristive neuromorphic devices face unresolved yield and variability challenges [43]; analog computing suffers from noise and device mismatch [44]; and photonic integration remains limited by immature fabrication and packaging [45]. Logic-based neural networks sidestep this barrier entirely, compiling directly to Boolean logic on mature, globally available transistor technology. This makes them uniquely positioned for near-term deployment at scale.

This leaves a critical gap between the recognized inadequacy of post-hoc optimization and the practical inaccessibility of non-von-Neumann hardware, motivating approaches that rethink model design while remaining grounded in mature, scalable computing platforms.

3 Logic-based Neural Networks

While the previous section examined compression-based optimizations of conventional neural networks, a fundamentally different approach to efficient inference is to design architectures that are efficient by construction, rather than compressing MAC-based models. This line of work reduces inference cost by replacing arithmetic-heavy operations with structurally simpler computations that better align with digital hardware.

Multiplication-free models such as binary neural networks [46–48] and bit-level summation architectures [49–51] exemplify this direction. However, removing multiplications alone does not eliminate abstraction overhead: these models still rely on operations that must be mapped onto gate-level logic, often incurring nontrivial cost in hardware.

Logic-based Neural Networks (LNN), often referred to as Weightless Neural Networks [52], represent a more radical departure from the MAC-based paradigm. They directly learn functions of lookup tables (LUTs) or explicit logic circuits, parameterizing the deployable model through structure rather than real-valued parameters. By learning the logic directly—rather than training a MAC-based

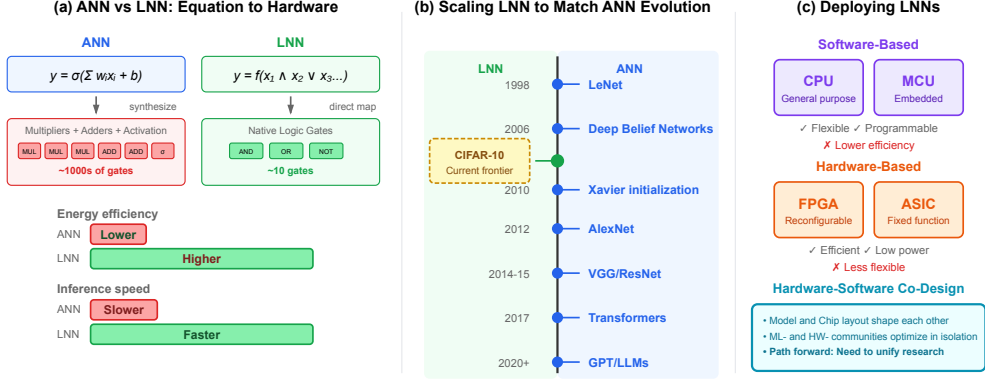


Figure 2: Logic-based neural networks and the path forward. **(a)** Why LNNs are efficient: conventional neural networks require weighted sums synthesized into multipliers and adders, demanding thousands of gates. LNNs formulate models directly as Boolean functions mapping onto native logic gates (AND, XOR, ...), yielding orders-of-magnitude fewer gates. **(b)** Current state: while ANNs have evolved from LeNet to LLMs over two decades, LNNs remain at an early stage, with the frontier limited to CIFAR-10-scale problems, roughly analogous to the pre-AlexNet era. **(c)** Unifying fragmented research: LNNs can be efficiently deployed as software or as hardware directly. Both require true hardware- software co-design for optimal results. But the community is split into ML- and hardware-focused research. We call to unify these efforts to accelerate LNN maturity.

surrogate and translating it post hoc—LNNs effectively discover compact task-specific circuits, distinguishing them from approaches like binary neural networks.

This architectural choice has significant implications for efficiency: inference reduces to LUT accesses and logic evaluation, with no multiplications and minimal data movement. As a result, LNNs are naturally suited to high-throughput, low-power deployment on ASICs and FPGAs, making them a strong candidate for hardware–software co-design in inference-constrained settings.

3.1 Approximative LUT Parametrization

Early approaches utilize single-layered perceptrons that approximate the input–output mapping of a LUT, enabling extremely fast and lightweight inference [53, 54]. However, this approximation incurs a fundamental expressivity gap: an n -input LUT has a VC-dimension of 2^n [55], whereas a single-layer perceptron with n inputs has VC-dimension $n + 1$, severely limiting its capacity to represent complex Boolean functions. Subsequent extensions have attempted to close this gap using higher-degree polynomial bases [56], multi-layered perceptrons [57–59] or gradient-boosted decision trees [60]. These methods show remarkable results on small-scale problems with inference in nano seconds and energy demands of nano Joule. However, they either remain insufficiently expressive or introduce exponential growth in training complexity as LUT size increases and face a fundamental tension between expressivity, training cost, and scalability.

3.2 Universal LUT Parametrization

To overcome the expressivity and scalability limitations, recent work has investigated universal representations of Boolean functions that admit gradient-based optimization. One prominent line of research represents Boolean functions as compositions of differentiable two-input logic gates, enabling end-to-end training of multi-layer architectures using backpropagation [19, 61]. While theoretically expressive, these approaches suffer from double-exponential parameter growth as function arity increases, limiting their practical scalability for deployment on modern FPGAs.

To directly address higher-degree LUTs, a parametrization is proposed in [20] that operates at the level of the LUT itself, achieving a logarithmic reduction in parameters compared to gate-based decompositions. Differentiability is obtained by relaxing the discrete addressing function to continuous values and approximating gradients through interpolation. More recently, alternative universal parametrization schemes for Boolean functions have been proposed [62–64], aiming to

eliminate both the double-exponential parameter growth of gate-based networks and the reliance on approximate gradients⁷.

Taken together, these approaches demonstrate that universal, differentiable parametrizations of LUTs are possible in principle, but existing methods face persistent trade-offs between expressivity, numerical stability, and scalability. Resolving these tensions remains a central challenge for logic-based neural architectures that aim to combine universality with hardware-efficient inference.

3.3 Hybrid Approaches

A central challenge for alternative inference paradigms is whether they can scale to the large, deeply layered models that dominate modern ML. In response, several recent works explore hybrid architectures that combine LNNs with conventional neural components, seeking to balance hardware efficiency with representational capacity [65–67]. These approaches typically retain standard deep learning structures where arithmetic intensity appears unavoidable, while selectively replacing specific submodules with logic- or LUT-based alternatives.

Recent transformer-scale experiments provide initial evidence that weightless components can be embedded into large architectures by substituting arithmetic-heavy feed-forward blocks with differentiable, LUT-based modules [67]. While such substitutions reduce reliance on multiply–accumulate operations and demonstrate functional compatibility with existing training pipelines, the resulting models remain only partially weightless and continue to depend on conventional dense layers for the majority of computation. Moreover, reported evaluations focus primarily on accuracy retention, leaving open questions regarding end-to-end inference cost, hardware realizability, and scalability under realistic deployment constraints. As such, current hybrid approaches should be viewed as exploratory stepping stones rather than definitive solutions to the efficiency challenges posed by large-scale models.

3.4 Current Limitations

Despite promising results on small-scale tasks, the scalability of current logic-based and weightless models remains constrained. Existing approaches face recurring challenges, including limited expressiveness [54, 56], reliance on approximate gradients [20], double-exponential parameter growth in gate-based constructions [19, 61], and discretization gaps between continuous training and discrete inference [22, 68]. Furthermore, finding robust encodings for continuous inputs remains an open challenge. Techniques such as stochastic relaxation and forward–backward alignment partially mitigate the discretization gap, but at the cost of added complexity and potential instability [22]. Advances in learnable connections [69–71] and learnable input encoding [72] come at the same drawbacks, albeit improving architectural flexibility and data efficiency, respectively.

Furthermore, training logic-based networks is often significantly slower than for conventional neural networks [61, 73]. While fair comparisons are hard to come by due to a lack of mature and optimized open source implementations, we estimate that LNNs are at least an order of magnitude computationally more expensive to train than other inference-optimized approaches.

Most work on LNNs has focused on modest-scale vision benchmarks such as CIFAR-10, with scaling explored primarily through increased width rather than depth, and end-to-end evaluations only exist for dense- and convolutional architectures. Extending LNNs to modern large-scale architectures and other data modalities hence remains an open challenge.

4 Our Position

We believe LNNs are a promising candidate to greatly increase computational efficiency of ML inference as they align model structure directly with the native capabilities of existing mature digital logic. At the same time, LNNs are far from being a mature or widely adopted alternative to conventional, linear-algebra-based models. The current ecosystem lacks robust software tooling, scalable training methodologies, standardized benchmarks, and compelling large-scale demonstrations. In the following, we outline why we believe these limitations are not fundamental, but rather reflect the

⁷As boolean logic is inherently non-differentiable, LNN gradients are technically always approximate. However, they can be exact w.r.t. the training objective when using a differentiable fuzzy logic surrogate.

early stage of the field—and how a coordinated community effort could enable LNNs to mature into a practical and impactful paradigm. Figure 2 sketches the relationship between conventional NNs and LNNs, and the where we see LNNs on a roadmap towards more wide-spread adoption.

4.1 Scaling to More Complex Models

In extending LNNs to realistic ML workloads, two challenges remain paramount: scalability with network depth and the incorporation of more expressive architectural primitives. Preliminary efforts have begun to extend LNNs beyond simple feedforward architectures. Early implementations of convolutional and recurrent LNNs exist [19, 74], but other important architectural families, most notably graph neural networks and transformers, remain largely unexplored in the logic-based setting, and no model exceeding $O(10)$ layers has been trained successfully.

We emphasize that these gaps should not be interpreted as fundamental roadblocks. Similar challenges arose during the early development of conventional neural networks. For example, scaling vision models from CIFAR-scale datasets to ImageNet required conceptual breakthroughs such as batch normalization and skip connections, culminating in architectures like ResNet [75]. These innovations fundamentally altered optimization dynamics and enabled training of very deep models. While initial attempts to bias LNNs toward identity mappings exist [19], true analogues of skip connections, normalization mechanisms, and other architectural stabilizers remain an open area of research.

There are already signs of progress. Recent reparameterizations reduce trainable parameters logarithmically, alleviating optimization burden [62, 72]. A small transformer has been deployed as a purely boolean circuit on an FPGA [76], and an LNN-based attention mechanism has been described [77]. Separately, hybrid architectures offer a practical deployment path as a stepping stone: computationally intensive MLP layers within transformers can for instance be replaced with LNN-based modules while retaining standard attention mechanisms [67].

Taken together, we believe that scaling LNNs to larger models and more complex architectures is feasible, but will require sustained research and development. This trajectory closely mirrors the historical evolution of matrix-based neural networks, where today’s standard techniques emerged only after years of community-wide experimentation.

4.2 Unifying Research Across Communities

A second major obstacle to progress is the fragmentation of the research landscape. LNNs have been studied both in the ML community and in the FPGA and digital design communities, but with markedly different emphases. The hardware-oriented community has focused on synthesis, deployment, and ultra-efficient inference on real devices, typically using very small benchmarks. In contrast, the ML community has explored training dynamics, scalability, and theoretical properties, often without considering realistic hardware constraints.

This separation has led to two fundamental problems. First, substantial duplication of effort has occurred, with identical or closely related ideas being developed independently under different names. Examples include reparameterizations of Boolean functions [62, 63] and the concept of learnable connections [20, 78]. Second, and more importantly, scaling and deployment are often studied in isolation, rather than in a true hardware–software co-design framework. This is also shown in Figure 1, where one can see that most papers pick either scaling or inference as their focus and rarely consider both.

In LNNs, architectural choices have immediate and intertwined consequences for both learning and implementation. For instance, restricting the fan-in of logic functions—such as limiting convolutional kernels to operate only on adjacent channels—can impose a stronger inductive bias toward locality, affecting model performance. At the same time, such restrictions reduce wiring complexity and enable more efficient physical implementations. Fully exploiting these trade-offs requires joint consideration of algorithmic, architectural, and hardware-level factors.

We therefore argue that closer integration between the ML and digital design communities is essential to unlock the full potential of LNNs. This paper is intended as a step in that direction, and we hope it will be read by researchers across both domains. More broadly, we advocate for institutional support mechanisms such as dedicated efficient ML tracks at major conferences. This explicitly encourages

cross-disciplinary work and evaluates contributions not only by accuracy, but also by reproducible measures of inference cost, latency, and energy.

Without such changes, the field risks continuing to reward marginal accuracy improvements while overlooking opportunities for orders-of-magnitude gains in efficiency—gains that are increasingly essential for sustainable, reliable, and widespread deployment of ML systems.

4.3 Lowering Barriers to Reproducible Deployment

The widespread adoption of common benchmark datasets has been instrumental in advancing ML by enabling reproducible and comparable evaluations across models and training methodologies. LNNs are typically benchmarked on these datasets as well, providing useful insight into achievable accuracy. However, efficiency must also be evaluated on hardware-centric metrics, including latency, energy consumption, and occupied silicon area.

In practice, such results are difficult to compare and reproduce, as the meta analysis in Figure 3 shows. It analyzes the accuracy and FPGA latency of different approaches on MNIST, the most wide-spread metrics. Inconsistencies arise from differences in hardware platforms and from varying conventions about whether the chip’s data input/output time is included in the measurements. We therefore advocate for the community to use benchmarks that jointly report task accuracy and inference performance on a clearly specified and shared set of hardware targets, enabling meaningful cross-comparison of approaches, similar to TinyML [5].

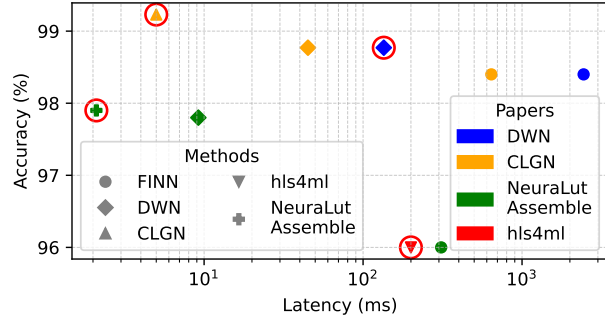


Figure 3: Overview of accuracy vs latency reports on MNIST from different papers. Red circles indicated self-evaluation.

Beyond comparability, deployment itself poses a significant challenge. Mapping a trained model onto a given hardware platform is a highly non-trivial process, with numerous opportunities for low-level optimization. Achieving competitive results typically requires substantial engineering effort tailored to each specific device. This creates a high barrier to entry, particularly for small research groups without dedicated hardware expertise. The situation is further exacerbated by the fact that many state-of-the-art approaches do not release their deployment pipelines, publishing only model architectures and training code.

To address these issues, we call for the development of a low-entry-barrier, open-source ecosystem that enables ML researchers to deploy, or at least realistically emulate the deployment of LNN models without extensive engineering expertise. A useful role model in this regard is hls4ml [79], which provides an accessible toolchain for synthesizing MAC-based neural networks for FPGA deployment. While such automatically generated designs may not fully exploit hardware capabilities and often leave room for manual optimization, the rapid iteration cycle and accessibility of hls4ml substantially lowered the barrier to research and enabled real-world deployments despite its non-optimal nature.

We believe that a similar approach is feasible and desirable for LNNs. Although existing LNN methods differ in how individual logic blocks are parameterized, approximated, and trained, they ultimately converge on a common abstraction: networks composed of low-level logic operations and their interconnections. This shared structure suggests that a unified deployment and benchmarking toolchain could serve a broad range of LNN approaches, fostering reproducibility, comparability, and wider adoption.

5 Call to Action

Efficient inference will not emerge from isolated advances; it requires coordinated effort across the ML ecosystem. To make this shift, inference cost must be treated as a first-class objective, on equal footing with accuracy. ML researchers should therefore design models with deployment constraints in mind, rather than relying solely on model compression.

To support meaningful comparison, hardware-relevant metrics such as latency, energy, and memory must be reported alongside task performance. This requires researchers to release open-source synthesis flows and reproducible deployment pipelines. End-to-end evaluations should be conducted on clearly specified and shared hardware targets to ensure comparability. Benchmark designers should then couple accuracy with inference cost under standardized deployment assumptions. Toolchain developers should provide low-barrier, open-source frameworks for training and deploying logic-based models, enabling wider adoption. Without these structural changes, opportunities for orders-of-magnitude efficiency gains will remain underexplored and unsolved.

6 Alternative Views

We acknowledge that there are several opposing views to this position.

LNNs will not scale to large architectures. Current differentiable LNNs achieve strong results on CIFAR-10 but struggle with CIFAR-100 and ImageNet [62]. Some of the main barriers come from a challenging gradient landscape and a high resource demand during training. As there are many promising developments in this field [22, 62, 67, 68, 72, 76, 77], we remain positive that this will change, but we acknowledge that it is possible that the development will slow down. A deliberate community effort will help pave the path to scalability.

The additional training and deployment cost outweigh inference gains. Logic-based models introduce significant upfront overhead: training is more resource-intensive, and their efficiency benefits often require specialized hardware. In many settings, this complexity is not justified, particularly when rapid iteration is critical. We acknowledge that LNNs are unlikely to replace frontier-scale models or applications with short product cycles. Instead, they are most compelling where strict constraints on power, latency, or chip area make conventional approaches infeasible, or where inference costs dominate over the model’s lifetime—for example, in stable “cognitive cores” within larger RAG-, reasoning-, or agentic systems.

Conventional methods become sufficiently efficient. GPU efficiency has improved dramatically: the NVIDIA GTX 580 used to train AlexNet in 2012 [80] delivered 6.5 GFLOPS/W in FP32⁸, while the H100 released twelve years later achieved 95.7 GFLOPS/W in FP32⁹—a roughly 25% annual improvement in FP32 efficiency. Combined with GPU-specific algorithmic advances such as FlashAttention [81] and compute-independent innovations such as speculative decoding [82], conventional hardware and software may continue improving fast enough that alternative computing paradigms become unnecessary. While such gains may reduce the need for alternatives, we expect demand to persist in regimes—however niche—where strict constraints or extreme scale make hardware-aligned solutions such as LNNs relevant.

7 Conclusion

The rapid growth of machine learning (ML) has made inference efficiency a defining constraint for both sustainability at scale and feasibility in resource-limited settings. While existing approaches have focused on optimizing within a linear-algebraic abstraction, we argue that the abstraction itself has become a bottleneck. By aligning model structure with the native capabilities of digital hardware, logic-based neural networks (LNNs) offer a path to fundamentally rethinking how inference is performed on mature transistor-based hardware. Although LNNs are not yet competitive across all regimes, their demonstrated efficiency gains and compatibility with existing fabrication technologies make them very attractive.

Realizing this potential will require a coordinated shift in research priorities, evaluation practices, and community structure. Progress depends not only on algorithmic advances, but also on investment in tooling, compiler support, hardware-realistic benchmarks, and cross-disciplinary collaboration between ML, and digital design. We argue that without such changes, the field risks continuing to trade marginal accuracy improvements for disproportionate increases in inference cost. By embracing LNNs as a first-class research direction, the community has an opportunity to pursue orders-of-magnitude gains in efficiency while leveraging mature, scalable transistor-based hardware.

⁸NVIDIA H100 GPU Datasheet. 2026-01-28

⁹NVIDIA GeForce GTX 580 Specs. 2026-01-28

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A ICML and NeurIPS topics over the years

Figure 1 was created by first collecting all the accepted abstracts from ICML and NeurIPS from 2015 to 2025. This was done by web-scraping the following sources: ICML: 2015-2016: Proceedings of Machine Learning Research [83, 84]. 2017-2025: ICML conference website (icml.cc/virtual/{year}/papers) NeurIPS: 2015-2024: NeurIPS paper archive (papers.nips.cc/paper_files/paper/{year}) 2025: NeurIPS 2025 conference website (<https://neurips.cc/virtual/2025/loc/san-diego/papers.html>)

The web-scraping was done with a custom pipeline following best practices and rules set out in the websites' robots.txt. To ensure responsible use and compliance with website terms of service, the scraper code is available from the authors upon reasonable request.

The abstracts were then passed to GPT-4o [85] to judge their content. The exact prompt was:

You are labeling research abstracts for a trend analysis. The labels are non-exclusive (multiple can be 1).

For each paper, output a JSON array with one object per paper (in order). Each object must include:

- id (int, 1-indexed)
- scaling_ml (0/1): The paper's main contribution involves scaling training/model capacity/data/compute (e.g., bigger models, scaling laws, distributed training, larger datasets, optimization for scaling, architecture changes primarily to enable scale).
- efficient_inference (0/1): The paper's main contribution improves inference-time efficiency (latency/throughput/memory/energy/cost at inference; compression, quantization, pruning, distillation, speculative decoding, serving systems, hardware acceleration for inference).
- llms (0/1): The paper centrally concerns large language models (LLMs) or language modeling as the main object of study.
- uncertain (0/1): Set to 1 only if classification is genuinely ambiguous from the abstract.
- justification (string): ≤ 20 words, mention the key phrase(s) that triggered the label(s).

Decision rules:

- Label based on the primary focus of the work, not incidental mentions.
- If the abstract clearly supports a label, set it to 1 even if other labels are also 1.
- If a paper is about speeding up training (not inference), that usually counts as "scaling_ml"=1, not "efficient_inference", unless inference is a main contribution too.

Output ONLY valid JSON (no markdown, no extra text).